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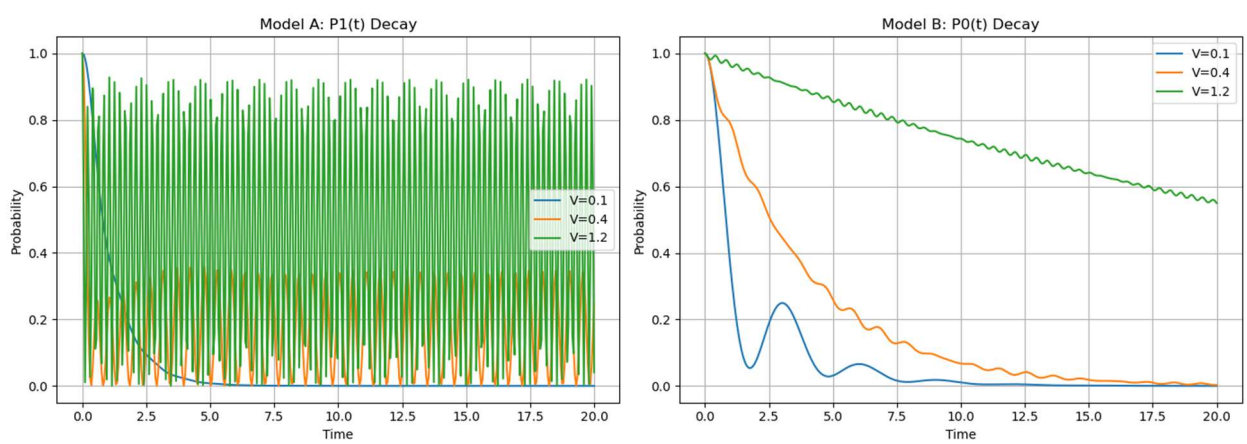
CHEM 5260 Midterm Project

Quantum mechanical models can provide deep insight into how macroscopic molecular processes occur at the microscopic level. Here, we analyzed two different models to simulate how a particle in an excited state can transfer into a continuum of states. This is analogous to modeling a single molecule in solution that can be promoted to an excited state by applying some perturbation, such as electromagnetic radiation, an applied potential, or an external magnetic field. The excited molecule could have some interaction with the solvent, represented in this model as the continuum of states. We are interested in investigating how the excited state evolves over time.

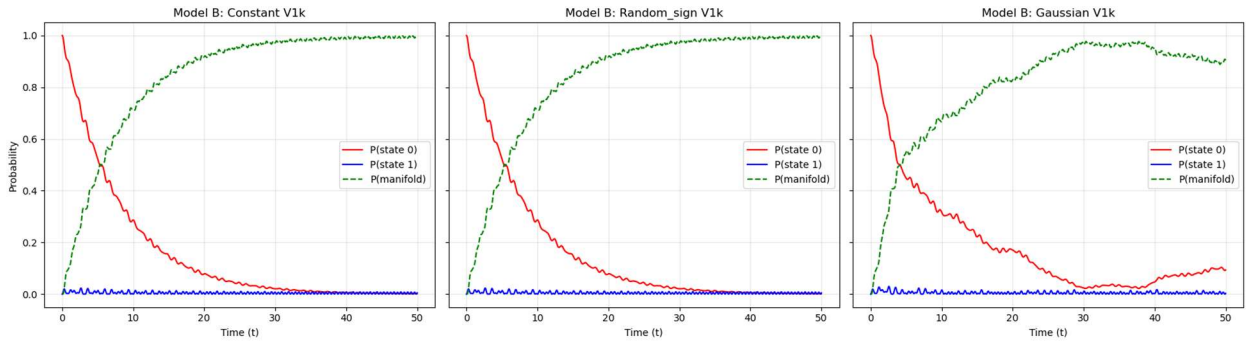
This research project seeks to investigate two models that simulate the molecular dynamics of a molecule in solution. The first model, Model A, contains a single state |1> that is coupled with N discrete energy levels. The second, Model B, has two states, one the source state (|0>), and another an intermediate state (|1>) between the source state and the continuum of states. Both models were tested to investigate the effect of couple strength (V), level density (N), bottlenecking (V_{01}), and energy spread (ΔE).

Coupling strength was first investigated, as it was suspected that there would be great limitations to the models correlated with changes in the coupling strength. Model A was analyzed for the probability of being in state |1>, and Model B was analyzed for the probability of being in state |0>. We observe that there is a clear threshold for V in model A, below which the function's behavior changes to oscillations.

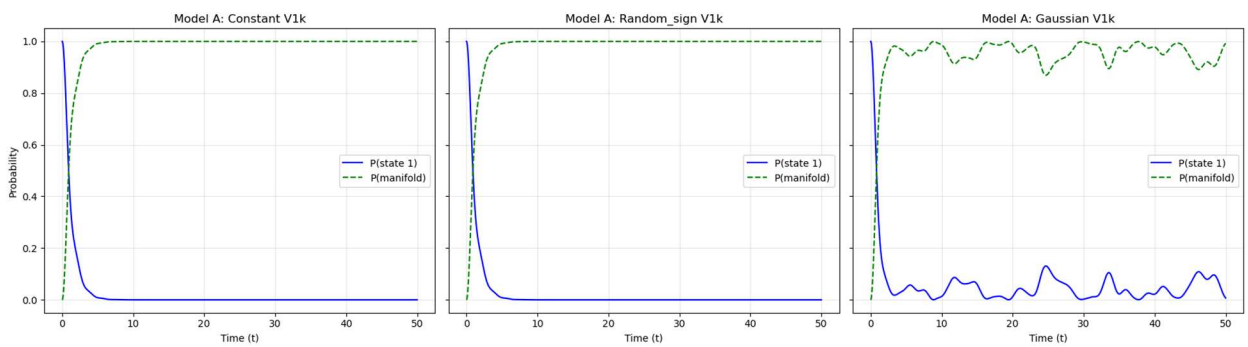


Continuing our investigation of Model B, now analyzing how the different states evolve over time as the distribution of coupling strength. These graphs demonstrate how changing V_{1k} affects the probability of finding the states. There is no significant difference between the constant value and a

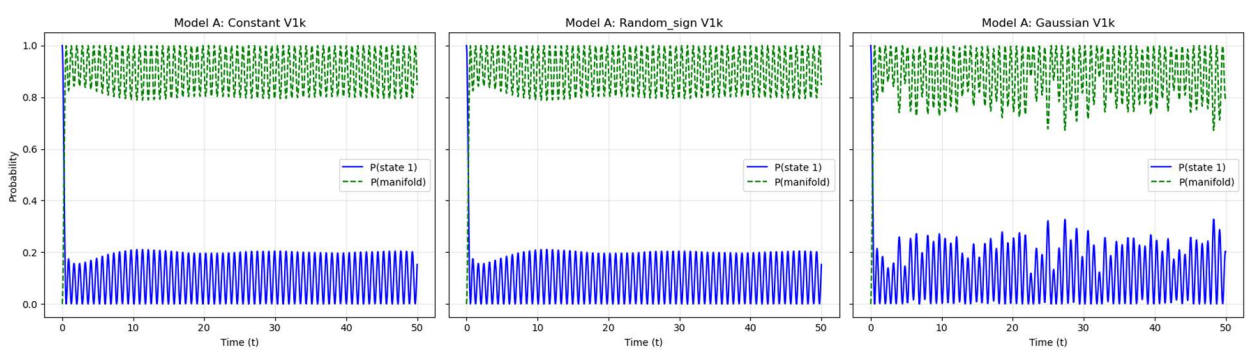
random sign for V_{1k}, indicating that the coupling's value has more influence than its magnitude (positive or negative). Even then, the general line shape in the Gaussian model is fairly similar to that of the other two graphs. It could be that model B depends less on V_{1k} than model A.



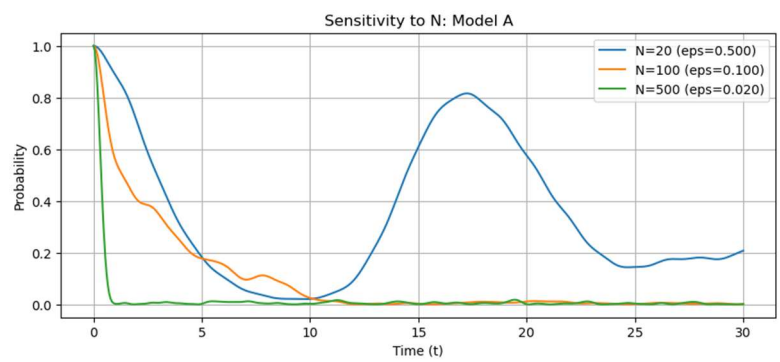
However, when the same analysis was performed with model A, we see a very similar trend. There are very few differences between the constant and random sign behavior. When V_{1k} is set to 0.1 (within the previously established limit), we observe the expected exponential decay. As with model B, choosing V_{1k} along a Gaussian, but within the magnitude of 0.1, results in very similar decay shape, but includes more fluctuations in the probability values.

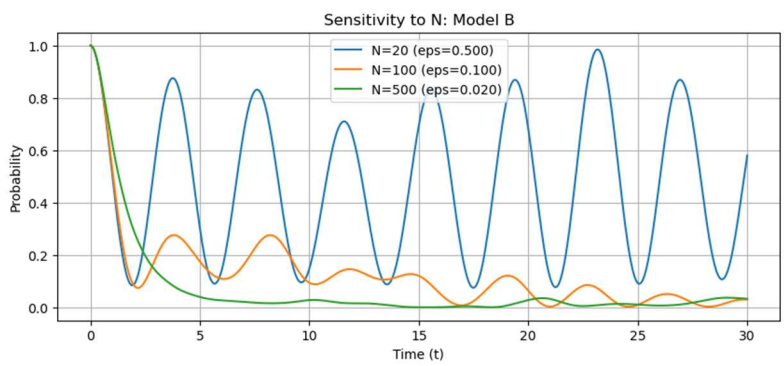


As described previously, if V_{1k} is set to a higher magnitude, in this case the magnitude of V_{1k} is 0.3, then the model fails and demonstrates oscillatory behavior.

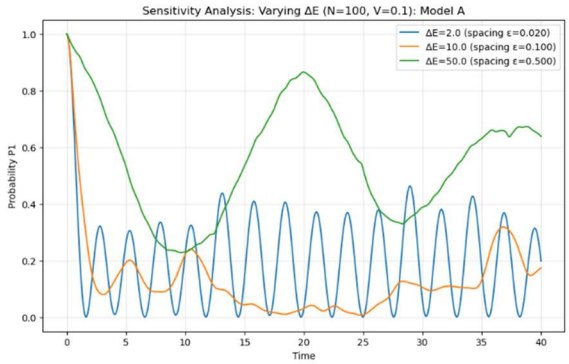


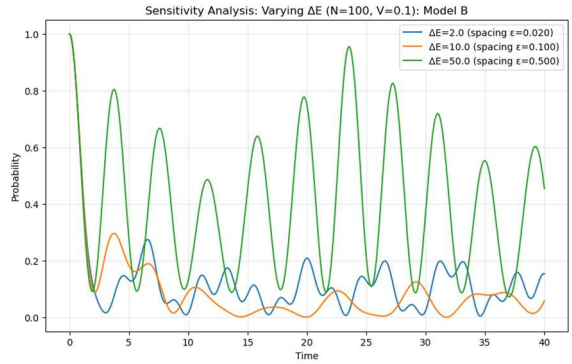
Now, we also investigated the sensitivity of both models to the value of N with a fixed value of V1k within the limit determined by previous results. At low values of N, Model A appears to follow the expected exponential decay only at early times. As the function continues to evolve, the behavior deviates as the probability of finding this state once again increased around 17. Although longer time scales were not explored, early behavior of the curve indicates that this oscillatory behavior may continue as t goes to infinity. With a large value of N, the probability peaks at early times and quickly decreases. This result makes sense because the electron will likely not remain in it's initial state when there are infinitely many energy levels available for transfer. Therefore, the probability will go to zero very quickly.



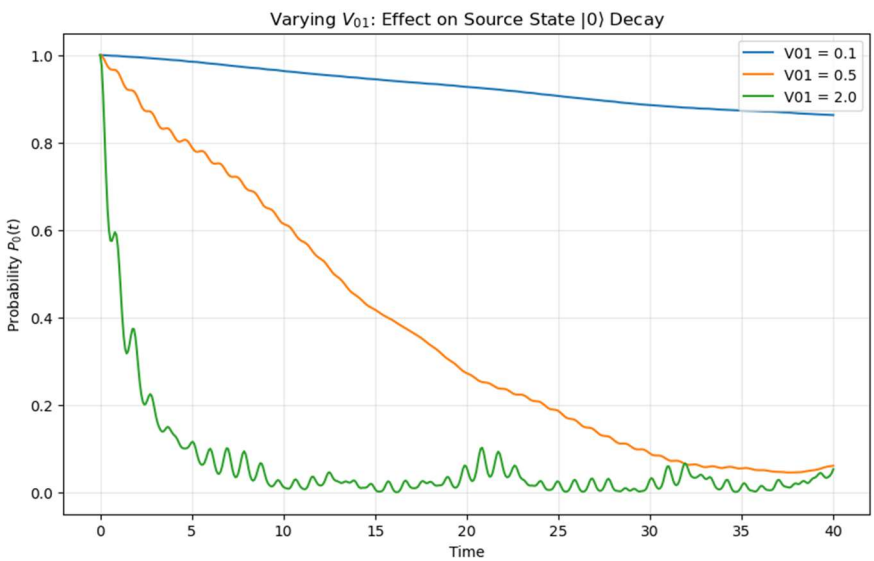


Interestingly, adjusting the value of N in model B causes a shift to oscillatory behavior, as well as a broader exponential decay. With low values of N, the probability of finding the electron in its initial state |0> oscillates, likely between |0> and |1>, as there are fewer states available in the continuum of states. In the case of N=100, we observed the expected exponential decay; however, there is still oscillations apparent in the line shape. As the number of continuous states approaches infinity, the oscillations disappear, leaving a smoother exponential decay.





Both models were also checked for sensitivity to energy distributions. ΔE represents the difference in energy between the starting state and the continuum of states. As expected, with a very large ΔE, the particle is more likely to remain in the initial state, as the energy difference is too much to overcome. There is no observable difference between an intermediate and large ΔE in Model B, possibly due to the fact that there is an intermediate state that can help facilitate energy transfer between the initial state and the continuum of states.



Lastly, Model B was investigated for the effect of coupling between states |0> and |1> on the evolution of the initial state. With low coupling values, the particle is likely to remain in the initial state. This greatly contrasts with large values of V_{01}, which causes the initial state to quickly decay. This is presumed to be because the initial state then evolves into state |1> and eventually the continuum of states.